EXPERIMENT 13 — Case Study on Low-Code Limitations

Aim: To study the limitations of low-code application development platforms using the EnergyWise – AI Building Energy Optimization project as a case study and identify suitable solutions to overcome these limitations.

Theory

Low-code development platforms allow applications to be developed using visual tools, predefined components and minimal programming. They simplify application development, but certain complex requirements may need additional programming, external integrations or specialized services.

In the EnergyWise project, Zoho Creator is used to manage building information, record energy consumption, generate reports and automate energy-related tasks.

The major limitations of low-code platforms include:

1. Limited customization: Complex application logic may require custom scripting.
2. Integration limitations: Connecting external sensors, APIs and third-party services may require additional configuration.
3. Advanced AI implementation: Energy forecasting and predictive analytics may require external machine-learning models.
4. Performance constraints: Applications with large volumes of energy readings may require optimized data handling and report design.
5. Platform dependency: Applications may depend on the features, pricing and restrictions of the selected platform.
6. Security and access management: Appropriate permissions and secure integration methods must be configured.

Procedure

1. Identify the main requirements of the EnergyWise application.

2. Study the features provided by Zoho Creator, including forms, reports, dashboards and workflows.
3. Compare the project requirements with the capabilities of the platform.
4. Identify requirements that may need external services or additional programming.
5. Suggest suitable solutions for the identified limitations.

Observation

Limitation	Proposed solution
Complex custom logic	Use Deluge scripting or external code.
External sensor integration	Connect through suitable APIs or middleware.
Advanced energy prediction	Integrate a suitable machine-learning model.
Large data volumes	Optimize reports, queries and data handling.
Platform dependency	Evaluate portability and supported export options.
Security risks	Apply access controls and secure credential management.

Result

The limitations of low-code application development were studied using the EnergyWise project as a case study. Suitable solutions, including custom scripting, API integration, machine-learning services, performance optimization and security controls, were identified.